«CUSTOMER NAME»
«GROUP COMPANIES»

Predictive Analytics · Service A

Predictive Analytics Pilot

Technical and Commercial Proposal

«USE CASE» on the Data of «CUSTOMER»: Data Readiness, Model, Validation and Measurement of Business Value

Proposal No: KOMTAS-«CUSTOMER ABBREVIATION»-PRA-A-«YEAR»-«SEQUENCE»

Proposal Date: «dd Month yyyy»

Validity: «proposal date + 1 month»

VARIABLE GLOSSARY — in this template only the fields inside « » change per customer. All other text is fixed.

Identity: «CUSTOMER» short customer name used in the text · «CUSTOMER NAME» full legal name (cover) · «GROUP COMPANIES» group companies in scope; delete the line for a single company · «CUSTOMER ABBREVIATION» short code in the proposal number · «YEAR» year · «SEQUENCE» proposal sequence number · «dd Month yyyy» proposal date · «proposal date + 1 month» end of validity · «price validity date» latest start date for which the fees stay valid

Context: «CUSTOMER CONTEXT» one paragraph: company or group, sector, customers or products concerned, where the data sits today, reason for the request · «TRIGGER» reason for the work (e.g. rising churn, stock-outs or overstock, low campaign efficiency, recent move of the data to BigQuery)

Use case: «FAMILY» use-case family: Retention & Churn, Demand & Supply, Pricing & Personalization, Risk & Collections or Industrial Operations · «USE CASE» use case from the catalogue in ANNEX-1 (e.g. asset churn prediction, demand forecasting, collection optimization) · «FAMILY — RETENTION & CHURN» block of the Retention & Churn family; keep it only if this family is in scope · «FAMILY — DEMAND & SUPPLY» block of the Demand & Supply family; keep it only if this family is in scope · «FAMILY — PRICING & PERSONALIZATION» block of the Pricing & Personalization family; keep it only if this family is in scope · «FAMILY — RISK & COLLECTIONS» block of the Risk & Collections family; keep it only if this family is in scope · «FAMILY — INDUSTRIAL OPERATIONS» block of the Industrial Operations family; keep it only if this family is in scope · «PLATFORM» delivery platform agreed with the customer: Google Cloud, Dataiku, Microsoft Azure or an open-source stack · «FREQUENCY» how often predictions or retraining are produced (e.g. monthly, weekly, daily) · «HORIZON» forecast horizon (e.g. 4 weeks, 13 weeks)

Scope: «BAND» sizing band S, M or L from the proposal input form

Commercial: «OPTIONAL» item kept or deleted by sales; if kept, delete the marker · «fee» fee excl. VAT · «amount» payment amount excl. VAT · «day rate» person-day rate (change requests) · «CITY» city where on-site meetings are free of charge

Values: «NUMBER» numeric value (count, threshold, days, weeks, months, hours) · «%» percentage

Annex: «RESULT» measured result of a reference project, only if measured and approved for use; otherwise delete the line

Contacts: «full name» contact name · «title» contact title · «e-mail / phone» contact details

Markers: of the paragraphs, rows and annex pages starting with «FAMILY — RETENTION & CHURN», «FAMILY — DEMAND & SUPPLY», «FAMILY — PRICING & PERSONALIZATION», «FAMILY — RISK & COLLECTIONS» and «FAMILY — INDUSTRIAL OPERATIONS», keep those of the family selected and delete the others; then delete the family markers. «OPTIONAL» items are kept or deleted by sales. The orange AX DOĞRULASIN and GÜNER DOĞRULASIN tags mark draft statements to be verified before use. FILL NOTE boxes are internal and are deleted before sending.

Contents

FILL NOTE — this template is for service A (Pilot) of the Predictive Analytics package. One use case of one family is built and tested on the data of «CUSTOMER»; nothing is put into production. Of the paragraphs, rows and annex pages starting with «FAMILY — RETENTION & CHURN», «FAMILY — DEMAND & SUPPLY», «FAMILY — PRICING & PERSONALIZATION», «FAMILY — RISK & COLLECTIONS» and «FAMILY — INDUSTRIAL OPERATIONS», keep those of the family selected and delete the others; then delete the remaining family markers, replace «FAMILY» and «USE CASE» with the family and the use case from the catalogue in ANNEX-1, and «PLATFORM» with the delivery platform.

Statements followed by an orange tag are draft assumptions of the package. AX DOĞRULASIN: verified with the Analytics team (use cases, data, metrics, durations, effort, thresholds, roles, references, acceptance). GÜNER DOĞRULASIN: verified with Cloud Solutions (cloud services, cloud architecture, consumption). After verification only the tag is deleted. No proposal is sent with a tag in it.

Order of completion: family and use case → platform (Section 2.4) → sizing values and band from the proposal input form (Table 5) → go / no-go thresholds (Table 7) → timeline (Section 4.2) → effort and fees (Section 9) → remaining placeholder fields. Margin, discount or incentive figures are never written into this proposal.

If «CUSTOMER» asks for production straight away, use template 4b (Production) only if a validated model of the same use case already exists; otherwise service B is not offered. «AX DOĞRULASIN»

1. Purpose

«CUSTOMER CONTEXT»

The aim of this proposal is to establish, before any investment in production, whether a «USE CASE» model built on the data of «CUSTOMER» reaches the agreed accuracy and whether its business value can be measured. The reason for the work is «TRIGGER».

The business question of the pilot is:

- «FAMILY — RETENTION & CHURN» Which customers are likely to leave, cancel or move to a lower segment in the next «NUMBER» months, and what drives it?
- «FAMILY — DEMAND & SUPPLY» How much will be sold or needed per product, store, site or period over the next «HORIZON»?
- «FAMILY — PRICING & PERSONALIZATION» Which offer, product or price fits each customer or segment, and through which contact?
- «FAMILY — RISK & COLLECTIONS» Which customers are likely to fall into delinquency or default, and which collection or credit action fits each level of risk?
- «FAMILY — INDUSTRIAL OPERATIONS» Which settings of the process keep quality and output while reducing energy use, deviations, defects or downtime?

Predictive Analytics builds prediction models on the data of the customer and puts their results into business systems. Its use cases are grouped into five families: Retention & Churn, Demand & Supply, Pricing & Personalization, Risk & Collections and Industrial Operations (ANNEX-1). Each use case follows the same lifecycle: data readiness check → feature set → model → validation → deployment → monitoring.

The package is offered as three services. This proposal covers service A.

Service	Focus	Result	Pricing
A — Pilot	One use case: data readiness, feature set, model, validation, value measurement	Validated model and go / no-go decision	Fixed fee
B — Production	Deployment pipeline, scheduled scoring, model registry, monitoring, integration	Model in operation, results in business systems	Fee per use case (separate proposal)
C — Run	Monitoring, retraining and improvement of the models in operation	Models kept accurate over time	Monthly fee (separate proposal)

Table 1 — Service structure. Service B is offered on the basis of the validated model of service A.

Service A is carried out in five phases:

- **P0 — Kick-off and data access:** scope, business question, data sources and access agreed; data made available in the pilot environment.
- **P1 — Data readiness check:** history, quality and label of the data checked against the needs of the use case.
- **P2 — Feature set and model:** feature set built, baseline and candidate models trained.
- **P3 — Validation and value measurement:** model validated on hold-out data; business value test designed and, where the period allows, first measured.

- **P4 — Results and go / no-go:** results presented; decision on production taken against the agreed thresholds.

Phase	Main Work	Duration
P0	Kick-off and data access	«NUMBER» weeks
P1	Data readiness check	«NUMBER» weeks
P2	Feature set and model	«NUMBER» weeks
P3	Validation and value measurement	«NUMBER» weeks
P4	Results and go / no-go decision	«NUMBER» weeks
Total		«NUMBER» weeks

Table 2 — Summary of the schedule. Band: «BAND».

At the end of the work, «CUSTOMER» knows whether its data is ready for the use case, has a validated model with documented features, a plan to measure business value with first results where available, and an estimate for production. The decision on production is taken on these results.

2. Scope

2.1 Use Case Selected

The pilot covers one use case: «USE CASE», from the «FAMILY» family. The use cases of each family are listed in ANNEX-1.

The output of the pilot depends on the family: «AX DOĞRULASIN»

Family	Business Question	Output of the Pilot
«FAMILY — RETENTION & CHURN» Retention & Churn	Which customers are likely to leave or downgrade, and why	Probability per customer and main drivers, on the pilot data
«FAMILY — DEMAND & SUPPLY» Demand & Supply	How much will be sold or needed per product, site and period	Forecasts per series and horizon on a back-test period
«FAMILY — PRICING & PERSONALIZATION» Pricing & Personalization	Which offer, product or price fits which customer	Offers, product lists, prices or segments per customer, evaluated offline or in a test design
«FAMILY — RISK & COLLECTIONS» Risk & Collections	Which customers are likely to become delinquent or default	Probability per customer with the main features; proposed contact or limit policy per risk level
«FAMILY — INDUSTRIAL OPERATIONS» Industrial Operations	Which process settings reduce energy, deviations or defects	Prediction of the process variable and recommended settings, evaluated on past or test periods

Table 3 — Families and pilot outputs. Delete the rows of the families not selected.

2.2 Scope Topics

The pilot covers the topics below. «AX DOĞRULASIN»

#	Topic	Content
1	Data readiness	History, completeness, quality and label of the data in scope, checked against the needs of the use case
2	Feature set	Features derived from the data sources, with their definitions and the time window they use
3	Model	A baseline model and at least one candidate model; the selected model with the reason for the choice
4	Validation	Technical metrics of the use case on hold-out data or on a back-test period
5	Value measurement	Control group or comparison design agreed with the business owner; first measurement where the period allows
6	Production plan	Target architecture, integrations, effort and duration of service B

Table 4 — Scope topics.

2.3 Scope Size

Sizing Input	Value	Source
Number of data sources	«NUMBER»	Statement of «CUSTOMER»
Months of history available	«NUMBER»	Statement of «CUSTOMER»
Volume: customers, products, time series, stores or process variables, depending on the use case	«NUMBER»	Statement of «CUSTOMER»
Planned prediction frequency	«FREQUENCY»	Statement of «CUSTOMER»
Business systems to receive the results in production	«NUMBER»	Statement of «CUSTOMER»
Data already on the delivery platform («PLATFORM»)	Yes / No	Statement of «CUSTOMER»

Table 5 — Values on which sizing is based. Band: «BAND». The values are statements and are checked in P1.

> *FILL NOTE — take the values and the band from the proposal input form (predictive-analytics-3a-pilot-girdi-formu). Draft band drivers:*
>
> *Data sources: S ≤3 · M 4–8 · L >8. Volume: customers S ≤100k · M 100k–1M · L >1M; time series S ≤10k · M 10k–100k · L >100k. Prediction frequency: S monthly · M weekly · L daily. Systems to integrate: S 1 · M 2–3 · L >3. Band rule: the highest band reached by at least three of the four drivers; the number of use cases is not a driver for the pilot.* «AX DOĞRULASIN»

2.4 Delivery Platform

The pilot is built on «PLATFORM». The lifecycle, the phases and the deliverables are the same on every platform; the platform is agreed with «CUSTOMER» before the start of the work. The options are:

Platform	Components Used in the Pilot
Google Cloud	BigQuery for the data, BigQuery ML or Vertex AI for the features and the models, results in BigQuery tables and Looker reports «GÜNER DOĞRULASIN»
Microsoft Azure	Azure Machine Learning, Azure Data Factory or Microsoft Fabric for data preparation, a Synapse or Fabric warehouse for the data
Dataiku	Data preparation, feature set, model training and evaluation in a Dataiku project connected to the data sources of «CUSTOMER» «AX DOĞRULASIN»
Open-source stack	Python-based open-source libraries on the infrastructure of «CUSTOMER»

Table 6 — Delivery platform options.

If the data in scope is not on the selected platform, it is loaded into the pilot environment as a one-off extract; a permanent data pipeline is not part of the pilot. «AX DOĞRULASIN»

2.5 Boundaries to Other Services

- **AI Readiness Assessment:** measures the readiness of an organisation for AI. This package builds a model.
- **ETL Modernization on Google Cloud and Data Modernization:** move the data platform. This package uses the data where it is, or a one-off extract.
- **Analytic CoE Enablement and MLOps platform:** build the analytics organisation and the model platform of the customer. This package delivers use cases.
- **Analytics platform migration:** moves existing models and data flows to a new platform (e.g. from SAS, R or Pentaho to Dataiku). This package builds new models.
- **Gemini Agent Kit:** covers generative AI agents and RAG / LLM work. This package covers prediction models.
- **AI Governance Readiness:** covers the governance of AI and models across the organisation.

3. Approach

3.1 Lifecycle

Each use case follows six steps: data readiness check → feature set → model → validation → deployment → monitoring. The pilot covers the first four steps and the design of the value measurement. Deployment is service B and monitoring in operation is service C.

3.2 Data Readiness Check

In P1 the data in scope is checked for: «AX DOĞRULASIN»

- **History:** enough months or years of data for the use case, without long gaps.
- **Coverage:** the customers, products, stores or assets in scope are present in all sources and can be joined.
- **Quality:** missing values, duplicates, outliers and changes of definition over time.
- **Label:** the outcome to be predicted is recorded and defined consistently.
- **Leakage:** no field contains information that would not be known at the time of the prediction.

The data required for the use case is agreed in P0 on the basis of the family page in ANNEX-1.

If the data does not meet the needs of the use case, this is reported at the end of P1 together with the options (continue with a reduced scope, complete the data first, or stop). The work continues only with the written approval of «CUSTOMER». «AX DOĞRULASIN»

3.3 Feature Set and Model

A baseline is set first: the current method of «CUSTOMER» where one exists, or a simple rule (e.g. last period's value, the most popular products). Candidate models are then trained and compared with the baseline on the same data. Each feature is documented with its source, its definition and the time window it uses, so that the same features can be produced in production. «AX DOĞRULASIN»

The importance and the direction of effect of the main features are reported to the business owner, so that the result of the model can be explained.

3.4 Validation

The model is validated on data that was not used in training: a hold-out period or group, or a back-test over past periods. The technical metrics by family are: «AX DOĞRULASIN»

- «FAMILY — RETENTION & CHURN» AUC or Gini and lift in the top decile.
- «FAMILY — DEMAND & SUPPLY» WAPE / MAPE and bias, against the current method of «CUSTOMER» on the same back-test period.
- «FAMILY — PRICING & PERSONALIZATION» precision@k and recall@k for offers and recommendations; for pricing, the fit of the estimated price elasticity on past periods.
- «FAMILY — RISK & COLLECTIONS» Gini on hold-out and out-of-time data.
- «FAMILY — INDUSTRIAL OPERATIONS» Prediction error of the process variable, or accuracy of the classification, on hold-out periods.

3.5 Value Measurement

The business value is measured separately from the technical metrics. In P3 a measurement design is agreed with the business owner: a control group that does not receive the action based on the model, or a comparison with the current method over the same period. The measurement period is «NUMBER» weeks. «AX DOĞRULASIN»

The business metric by family is: «AX DOĞRULASIN»

- «FAMILY — RETENTION & CHURN» Difference in churn or segment downgrade between the customers targeted with the model and a control group.
- «FAMILY — DEMAND & SUPPLY» Change in stock-outs, overstock or stock holding cost compared with the current method, estimated on the back-test period.
- «FAMILY — PRICING & PERSONALIZATION» Revenue, conversion or click-through rate of a test group against a control group or the current offer or price.
- «FAMILY — RISK & COLLECTIONS» Transition into delinquency against a control group, or credit approval rate at the same risk level.
- «FAMILY — INDUSTRIAL OPERATIONS» Energy consumption, deviation, quality or yield of test units or periods against control units or periods.

If the measurement period does not fit into the pilot, the measurement plan is delivered in P3 and the measurement continues under the responsibility of «CUSTOMER» or in service B. «AX DOĞRULASIN»

The business result depends on the actions of «CUSTOMER» based on the model (e.g. campaigns, orders, prices, contact policy, process settings). It is measured and reported; it is not guaranteed by this proposal.

3.6 Go / No-Go Decision

At the end of P4 the decision on production is taken against the criteria below, agreed in P0.

Criterion	Threshold	Source
Technical metric of the use case on hold-out data or back-test	«NUMBER» «AX DOĞRULASIN»	Validation report
Improvement against the baseline	«NUMBER»	Validation report
Data ready for regular production runs	Yes / No	Data readiness report
Value measurement design agreed with the business owner	Yes / No	Value measurement plan

Table 7 — Go / no-go criteria.

A no-go decision is a result of the pilot. In that case the report states the reason and what would need to change (e.g. more history, a different label, additional sources).

4. Activities and Timeline

4.1 Phases

The work is carried out in five phases. Each phase has a written exit criterion. «AX DOĞRULASIN»

Phase	Activities	Exit Criterion
P0 Kick-off and data access	Kick-off, business question, data sources, access, go / no-go thresholds	Scope, data access and thresholds agreed in writing
P1 Data readiness check	Profiling of the data, check of history, coverage, quality, label and leakage	Data readiness report accepted; decision to continue
P2 Feature set and model	Feature set, baseline, candidate models, model selection	Selected model and feature definitions documented
P3 Validation and value measurement	Validation on hold-out data or back-test, measurement design, first measurement where possible	Validation report and value measurement plan published
P4 Results and go / no-go	Results presentation, production plan and estimate, decision	Go / no-go decision recorded

Table 8 — Phases and exit criteria.

4.2 Timeline

FILL NOTE — the duration follows the band in Section 9. Draft distribution of the weeks for band M (7 weeks): P0 1 · P1 1.5 · P2 2 · P3 1.5 · P4 1 weeks. «AX DOĞRULASIN»

Phase	Work Step	Weeks
P0	Kick-off, business question, data access	«NUMBER»
P1	Data readiness check	«NUMBER»
P2	Feature set, baseline and candidate models	«NUMBER»
P3	Validation, measurement design, first measurement	«NUMBER»
P4	Results, production plan, go / no-go	«NUMBER»
	Total	«NUMBER»

Table 9 — Distribution of the weeks. Band: «BAND».

The timeline applies provided the assumptions in Section 7 hold. Delays in data access or in the delivery of the data move the timeline by the same amount; they cannot be compensated by adding people.

5. Deliverables and Acceptance

5.1 Deliverables

Documents are delivered in Turkish, in editable format (Microsoft Word / Excel / PowerPoint) together with a PDF copy. The code and the model remain in the environment of «CUSTOMER».

#	Deliverable	Phase	Content
D1	Kick-off note and data access plan	P0	Business question, data sources, access, go / no-go thresholds
D2	Data readiness report	P1	History, coverage, quality, label and leakage per source; gaps and options
D3	Feature and model definitions	P2	Feature definitions, baseline, candidate models, selected model and reason
D4	Validation report	P3	Technical metrics on hold-out data or back-test, comparison with the baseline
D5	Value measurement plan and first results	P3	Control group or comparison design, measurement period, business metric, first results where available «AX DOĞRULASIN»
D6	Production plan and estimate	P4	Target architecture, integrations, effort and duration of service B
D7	Results presentation and go / no-go record	P4	Results against Table 7 and the decision

Table 10 — Deliverables.

5.2 Acceptance

The technical metric and the business metric are accepted separately. The technical metric is measured on hold-out data or a back-test; the business metric is measured and reported, not guaranteed.

Deliverable	Acceptance Criterion
Data readiness report	Every source in scope assessed for history, coverage, quality and label; gaps listed with options «AX DOĞRULASIN»
Validation report	Technical metric of the use case computed on hold-out data or back-test and compared with the baseline and with the threshold in Table 7
Value measurement plan	Design, period and business metric agreed with the business owner of «CUSTOMER»
Production plan and estimate	Architecture, integrations, effort and duration given together with the assumptions they rest on

Table 11 — Acceptance criteria.

Each deliverable is reviewed by «CUSTOMER» within «NUMBER» working days of delivery. If no written objection is received in that period, the deliverable is deemed accepted. A no-go result does not prevent the acceptance of the deliverables.

6. Customer Responsibilities

«CUSTOMER» is responsible for the items below. A delay in any of them moves the timeline by the same amount.

- Appoint a project coordinator who is available throughout the work.
- Make the data in scope available in the pilot environment within «NUMBER» working days from the start of P0.
- Ensure the legal basis under KVKK for using customer data in the model; share personal data only where needed, with pseudonymised identifiers where possible. «AX DOĞRULASIN»
- Agree the label, the business metric and the go / no-go thresholds with Komtaş in P0.
- Make the business owner available to agree the value measurement design and, where it is run, to carry out the action for the target group.
- Name the decision maker who attends the results presentation in P4.

The participation expected from «CUSTOMER» is: «AX DOĞRULASIN»

Role	Expected Contribution	Participation
Project sponsor	Approval of scope, go / no-go decision	P0, P4
Business owner (e.g. marketing, supply chain, risk, collections or production)	Business question, label, business metric, value measurement	P0, P3, P4
Data owner	Data sources, access, meaning of the fields, data questions	P0 – P2
IT	Access to the data and to the pilot environment; target systems for production	P0, P4

Table 12 — Expected participation of «CUSTOMER». Total expected effort is «NUMBER» person-days.

7. Assumptions and Out of Scope

7.1 Assumptions

The duration, effort and fee in this proposal are based on the assumptions below. If they do not hold, the impact is assessed through a change request.

- The work starts within «NUMBER» working days of contract signature.
- The values in Table 5 have been stated by «CUSTOMER». If the data readiness check in P1 indicates a different band, duration and effort are reassessed at the end of P1. «AX DOĞRULASIN»
- The pilot environment on «PLATFORM» is provided by «CUSTOMER» or set up in its cloud account; cloud consumption and platform licences are paid by «CUSTOMER». «GÜNER DOĞRULASIN»
- The pilot uses a one-off extract of the data (Section 2.4); the data is not refreshed during the pilot.
- The work is carried out remotely; the kick-off and the results presentation can be held on site in «CITY». «AX DOĞRULASIN»
- The working language and the language of the deliverables is Turkish.

7.2 Out of Scope

The topics below are outside the scope of this proposal. If requested, they are assessed and proposed separately. «AX DOĞRULASIN»

- Deployment of the model to production, scheduled scoring and integration to business systems; these are the subject of service B.
- Monitoring and retraining in operation; these are the subject of service C.
- Use cases other than the one selected in Section 2.1.
- Migration of the data platform or of the analytics platform.
- Cleansing of data in the source systems.
- Execution of campaigns, orders, price changes, collection actions or process changes based on the model.
- A guaranteed business result.
- Real-time (online) scoring, unless stated in Section 2.
- Cloud consumption and licence costs.

8. Team

8.1 Komtaş Team

Role	Responsibility	Participation
Data Scientist (lead)	Method, feature set, models, validation, results	P0 – P4 «AX DOĞRULASIN»
ML Engineer	Pilot environment, training runs on «PLATFORM», production plan	P0 – P4, part time
Data Engineer	Data access, extract, profiling, data preparation	P0 – P2
Business Analyst / Engagement Lead	Business question, label, value measurement design, plan and phase exits	P0 – P4, part time

Table 13 — Komtaş team and roles.

- If a change in the team is needed, the replacement has at least equivalent qualifications and is notified to «CUSTOMER» in advance.

8.2 Governance

- Weekly progress meeting on data access, open items and timeline.
- Review and written approval at the end of each phase.
- The data readiness results and the first model results are shared as interim results in P1 and P2; the results are not presented all at once at the end.
- Any issue that affects the timeline or the scope is reported in writing in the week it is found.

8.3 Risks

Risk	Impact	Mitigation
Insufficient history or long gaps in the data	Accuracy, go / no-go	History checked in P1; options reported before the model work starts
Unclear or changing label (e.g. churn or default definition)	Validity of the model	Label agreed in writing with the business owner in P0
Business does not act on the results	Value cannot be measured	Value measurement design and owner agreed in P3
Delay in data access or in the KVKK approval	Timeline	Data access plan and dates fixed in P0
Information in the data that would not be known at prediction time	Over-optimistic metrics	Leakage check in P1 and P3
Stated scale differs from the data	Band, duration	Band confirmed at the end of P1; the difference is reported in writing

Table 14 — Risks.

Any request that affects the scope, the timeline or the fee is handled through a written change request. It is not carried out before its impact and fee are approved.

9. Effort and Fees

> *FILL NOTE — take the band, the duration and the effort from the proposal input form (predictive-analytics-3a-pilot-girdi-formu). Fees are entered by sales. Draft values:*
>
> *Duration and Komtaş effort: S 6 weeks / ~45 pd · M 7 weeks / ~60 pd · L 8 weeks / ~75 pd. «AX DOĞRULASIN»*
>
> *The values are not based on recorded effort and are updated with the data of the Analytics team. Delete this box before sending.*

Service A is offered at a fixed fee for one use case. The fee is based on the sizing values in Table 5 and the assumptions in Section 7.

Item	Person-Days	Fee (excl. VAT)
P0 — Kick-off and data access	«NUMBER»	«fee»
P1 — Data readiness check	«NUMBER»	«fee»
P2 — Feature set and model	«NUMBER»	«fee»
P3 — Validation and value measurement	«NUMBER»	«fee»
P4 — Results and go / no-go	«NUMBER»	«fee»
Total	«NUMBER»	«fee»
Change requests	Per person-day	«day rate»

Table 15 — Fees. Band: «BAND». All fees exclude VAT.

Payment Item	Rate	Amount (excl. VAT)
Contract signature	«%»	«amount»
Acceptance of the deliverables (P4)	«%»	«amount»

Table 16 — Payment schedule. All fees exclude VAT.

The fee does not include:

- Deployment and operation of the model (services B and C).
- Cloud consumption and licence costs.
- Travel and accommodation outside «CITY».
- Additional work approved through change requests.

10. Next Step

The results of the pilot are the basis of the next steps. The usual follow-on work is:

Need	Follow-On Work
Putting the model into operation	Service B Production: deployment pipeline, scheduled scoring, integration to business systems
Keeping the model accurate	Service C Run: monitoring, retraining, monthly report
Other business questions	Pilot of a further use case from the same or another family (ANNEX-1)
Data in on-premises ETL tools or warehouses	ETL Modernization on Google Cloud, Data Modernization
Analytics organisation and model platform	Analytic CoE Enablement Program, MLOps platform
Governance of the models	AI Governance Readiness
Cloud consumption	Purchase of Google Cloud or Microsoft Azure consumption through Komtaş

Table 17 — Follow-on work.

10.1 Production and Run

After a go decision, service B is proposed on the basis of the production plan of this pilot. Service C can start once the model is in operation; it is offered with a monthly capacity of hours.

10.2 Further Use Cases

The other use cases follow the same lifecycle and can use the same platform. A pilot of a further use case can reuse the data access and the environment of this pilot.

10.3 Data Platform

Where the data sits in on-premises ETL tools or warehouses, the migration of the data platform (e.g. to Google Cloud) and the model work can be planned together. «GÜNER DOĞRULASIN»

10.4 Consumption

Komtaş is a Google Cloud Premier partner and has completed its partnership with Microsoft Azure. The cloud consumption of «CUSTOMER» can be purchased through Komtaş; the commercial terms are set out in a separate proposal. «GÜNER DOĞRULASIN»

11. Commercial Terms

11.1 General Terms

- All fees in this proposal are stated in Turkish lira and exclude VAT. Statutory taxes, duties and charges are not included in the fees.
- The proposal is valid until «proposal date + 1 month».
- The fees are valid on condition that the work starts by «price validity date». If the start moves beyond that date for any reason, the fees and pricing terms are reassessed by mutual agreement in the light of the economic and financial conditions at the time of the work.
- This work is offered at a fixed fee, based on the sizing values in Table 5 and the assumptions in Section 7.
- The business result of the model is measured and reported; it is not guaranteed.
- «OPTIONAL» If the contract for service B is signed within «NUMBER» months of the acceptance of this work, the fee of this work is deducted from the fee of service B.
- The start date of the work is agreed with «CUSTOMER». All activities are carried out mainly online; on-site meetings take place within «CITY».
- Payment follows the payment schedule in Section 9. Invoices are payable within «NUMBER» days of the invoice date.
- Scope changes are managed through a written change request; no change is implemented before its impact analysis is approved.
- The parties keep confidential all information they access within the scope of the work. A separate non-disclosure agreement is signed where needed.
- The rights of use of the deliverables belong to «CUSTOMER»; Komtaş retains its intellectual property rights in its methodology, templates and general know-how.
- This proposal forms an integral annex to the contract to be signed by the parties. In the event of a conflict between the contract and the proposal, the contract prevails.

11.2 Proposal Contact Details

Questions on this proposal and its scope can be addressed to the contacts below.

Area of Responsibility	Full Name	Title	Contact
Commercial	«full name»	«title»	«e-mail / phone»
Technical	«full name»	«title»	«e-mail / phone»

11.3 Proposal Annexes

This proposal is submitted with the following annexes:

- ANNEX-1 — Use-Case Family Pages
- ANNEX-2 — About Komtaş

ANNEX-1 — Use-Case Family Pages

> *FILL NOTE — keep only the page of the family selected in Section 2.1 and delete the other four.*

The use cases of Predictive Analytics are grouped into five families. Every use case follows the same lifecycle: data readiness check → feature set → model → validation → deployment → monitoring. The lifecycle does not change with the delivery platform. The page of each family in scope is given below; field references and measured results are listed in Table 24 (ANNEX-2).

«FAMILY — RETENTION & CHURN» Retention & Churn

The use cases, data, outputs and success metrics of the Retention & Churn family are: «AX DOĞRULASIN»

Item	Content
Use cases	Path to Churn (action on the critical steps before the decision to churn); "balance" churners (incoming funds withdrawn after a few days); asset churn and segment downgrade (banking); subscription and membership churn; customer satisfaction index.
Problem	Customers leave, cancel or move their assets; retention actions reach them late or are spread over all customers at the same cost.
Typical sectors	Banking, fitness and subscription services, telecommunications, retail loyalty programmes, insurance.
Data	At least 12–24 months of customer, product or contract, transaction, visit or usage, campaign and complaint data; in banking, the history of assets and segments.
Label	Churn, cancellation or segment downgrade as defined with the business owner (e.g. no renewal, or no visit for «NUMBER» days).
Output	Probability per customer and the main drivers behind it, refreshed «FREQUENCY»; critical events passed to sales or marketing as alert-action scenarios.
Success metrics	AUC or Gini and lift in the top decile on hold-out data. Business metric: difference in churn or segment downgrade between the targeted customers and a control group.

Table 18 — Retention & Churn family.

«FAMILY — DEMAND & SUPPLY» Demand & Supply

The use cases, data, outputs and success metrics of the Demand & Supply family are: «AX DOĞRULASIN»

Item	Content
Use cases	Demand forecasting (retail, manufacturing); inventory productivity analysis; visitor and sales prediction (shopping malls); schedule optimization.
Problem	Stock-outs, overstock and planning errors arise when the quantities ordered, produced or allocated do not follow demand.
Typical sectors	Retail, e-commerce, FMCG, manufacturing, distribution, shopping mall management.
Data	At least 2–3 years of sales or demand at product–store or product–site and day or week level, prices and promotions, stock levels and stock-outs, calendar and holidays; external data where relevant (e.g. weather, price indices, visitor counts).
Output	Forecast per series and horizon («HORIZON»), refreshed «FREQUENCY»; scenario runs where the business needs them.
Success metrics	WAPE / MAPE and bias against the current method of «CUSTOMER» on a back-test period. Business metric: change in stock-outs, overstock or stock holding cost.

Table 19 — Demand & Supply family.

«FAMILY — PRICING & PERSONALIZATION» Pricing & Personalization

The use cases, data, outputs and success metrics of the Pricing & Personalization family are: «AX DOĞRULASIN»

Item	Content
Use cases	Next best offer; product bundling; recommendation; customer lifetime value; deposit pricing; credit pricing; list price and markdown optimization; dynamic pricing; customer segmentation; customer contact policy; onboarding analytics.
Problem	Offers, prices and contacts are the same for all customers or follow fixed rules; they do not follow the response of each customer or segment.
Typical sectors	Banking, retail, e-commerce, spare parts and distribution, media.
Data	At least 6–12 months of transactions, offers and responses, prices, discounts or negotiation records, the product catalogue and hierarchy and customer attributes; for recommendation, interactions (views, baskets, purchases).
Output	Offer, product list, price or segment per customer or per product group and customer segment, refreshed «FREQUENCY»; for pricing, the price elasticity per product group and customer segment.
Success metrics	precision@k and recall@k offline; click-through rate, conversion or revenue in an A/B test against the current offer or price. Business metric: revenue or conversion of a test group against a control group.

Table 20 — Pricing & Personalization family.

«FAMILY — RISK & COLLECTIONS» Risk & Collections

The use cases, data, outputs and success metrics of the Risk & Collections family are: «AX DOĞRULASIN»

Item	Content
Use cases	Collection analytics and collection optimization; credit risk model enhancement, including graph analytics on relational data; dynamic client limit; risk analytics.
Problem	Collection contacts and credit decisions apply the same rules to customers with different levels of risk; existing models do not use the relations between customers.
Typical sectors	Banking and other lenders.
Data	Credit, repayment and delinquency history, limits and their use, customer and product data, contact history of the collection channels (call centre, IVR, SMS / push); for graph analytics, payments, transfers and links between customers (e.g. partnership, guarantorship).
Label	Delinquency within the next «NUMBER» days, or a default definition agreed with risk management.
Output	Probability of delinquency or default per customer, with the importance and direction of effect of the main features; contact policy or limit distribution per risk level.
Success metrics	Gini on hold-out and out-of-time data. Business metric: transition into the first delinquency cycle against a control group, or credit approval rate at the same NPL ratio.

Table 21 — Risk & Collections family.

«FAMILY — INDUSTRIAL OPERATIONS» Industrial Operations

The use cases, data, outputs and success metrics of the Industrial Operations family are: «AX DOĞRULASIN»

Item	Content
Use cases	Energy optimization (HVAC systems, furnaces); predictive quality and quality classification; production yield optimization; predictive maintenance; process unit prediction; fixed asset utilization.
Problem	Processes are run with settings chosen by experience; energy use, deviations, defects and downtime are seen only after they occur.
Typical sectors	Manufacturing (e.g. glass, pharmaceuticals), refining and petrochemicals, other energy-intensive production.
Data	Sensor and IoT time series of the process (temperatures, flows, energy, set values), laboratory samples, quality, maintenance and production records; controllable and uncontrollable factors identified with the process engineers.
Output	Prediction of the process variable (e.g. energy, temperature, quality class, yield) and recommended settings within the operating constraints, based on ML-guided simulation.
Success metrics	Prediction error of the process variable or accuracy of the classification on hold-out periods. Business metric: energy consumption, deviation, quality or yield of test units or periods against control units or periods.

Table 22 — Industrial Operations family.

ANNEX-2 — About Komtaş

> *FILL NOTE — the figures in Table 23 and in the bullets below come from the proposals of 2026. Check them against the current corporate profile before sending.*

Komtaş is an independent information technology company that has specialised in enterprise data management in Türkiye since 1989. More than 35 years of focus on a single field has given it an end-to-end view that follows data from the moment it is created to the moment a decision is taken.

Indicator	Value
Founded	1989 — more than 35 years of continuous data-focused activity
Specialist staff	250+ data engineers, architects, consultants and analysts
Offices	Istanbul, Ankara, Izmir, Dubai
Strategic partnerships	10+ global technology vendors
Focus area	Data governance, data architecture, analytics, artificial intelligence and AI agent based solutions
Positioning	"From Data to Agents" — integrated capability from data to AI agents

Table 23 — Komtaş corporate profile.

Service Capabilities

The service portfolio of Komtaş covers the whole life cycle of a data initiative:

- **Data Governance and Data Management:** data governance maturity assessment, framework and organisation design, policy and process design, data catalogue, business glossary, data quality management, master data management, data privacy and classification.
- **Data Architecture and Engineering:** data warehouse and data lake architectures, lakehouse design, data integration and streaming architectures, cloud and on-premises modernisation.
- **Analytics and Artificial Intelligence:** enterprise reporting, advanced analytics, machine learning models, generative AI applications and AI governance.
- **Managed Services and Support:** P1 / P2 / P3 support for production systems, capacity and performance management, continuous improvement services.
- **Training and Capability Building:** product training, role-based data governance training, train-the-trainer programmes.

Platforms and Partnerships

Komtaş is a Google Cloud Premier partner. Its partnership with Microsoft Azure has been completed.

The use cases of this package are delivered on the platform agreed with the customer: Google Cloud, Dataiku, Microsoft Azure or an open-source stack. The lifecycle and the services are the same on every platform.

Predictive Analytics References

Reference projects of Komtaş in the five families of this package, with the results measured in them, are listed below. Reference calls with the customers concerned can be arranged on request. «AX DOĞRULASIN»

FILL NOTE — keep the rows of the family in scope or all rows. Do not change the results and do not add customer names, contact persons or results that are not approved for use in proposals.

Family	Use Case	Organisation	Result
Retention & Churn	Asset churn prediction (segment downgrade)	Odeabank	11% lower segment downgrade in the test group than in the control group
Demand & Supply	Demand prediction	A tyre manufacturer	7% lower stock holding cost for 7 tyre types
Demand & Supply	Visitor and sales prediction	A shopping mall management company	Forecasts for 350+ stores in 2 malls with 1.6 million visitors
Pricing & Personalization	List price and markdown optimization	A spare-parts seller	+4% revenue at the same conversion level in the test group; price elasticity for 30,000+ SKUs; 12 A/B test scenarios
Risk & Collections	Collection optimization	Albaraka Banking Group, TEB	3–7% reduction in transition into the first delinquency cycle; Gini 0.86; explainable AI; optimized contact policy
Risk & Collections	Credit risk model enhancement with graph analytics	QNB Finansbank	+4.5% credit approval rate at the same NPL ratio; Gini 0.78–0.81
Industrial Operations	Energy optimization in HVAC systems	Eczacıbaşı Monrol	2.3% lower energy consumption in the test units than in the control group
Industrial Operations	Furnace energy optimization, furnace temperature control, glass quality classification, production yield optimization	Şişecam	«RESULT»
Industrial Operations	Continuous catalytic reformer (CCR) unit prediction	An integrated oil and petrochemical group	Qualitative: less error-prone operation of the unit and more control over the process

Table 24 — Predictive analytics references of Komtaş.

Komtaş has also built a subscription churn model for a fitness / subscription company (Retention & Churn) and a demand forecasting model for a retail company (Demand & Supply). Result: «RESULT». «GÜNER DOĞRULASIN»

Why Komtaş

- **Focus:** Komtaş has worked only in the data field for more than 35 years, with more than 250 specialists and 4 offices.
- **One method for five families:** retention, demand, pricing, risk and industrial use cases follow the same steps, from the data readiness check to monitoring.
- **Measured results:** the references in Table 24 give the results measured in the work, where they are available.
- **Data and model together:** the same organisation prepares the data, builds the models and operates them, on the platform of the customer.
- **From pilot to operation:** pilot, production, monthly operation and cloud consumption are provided by the same organisation (Section 10).